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Backend Development with Node.js, Express, and Mongo

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CLASS : 5BTCSIOT

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Practical Question Paper — Set 2

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Hospital Patient System

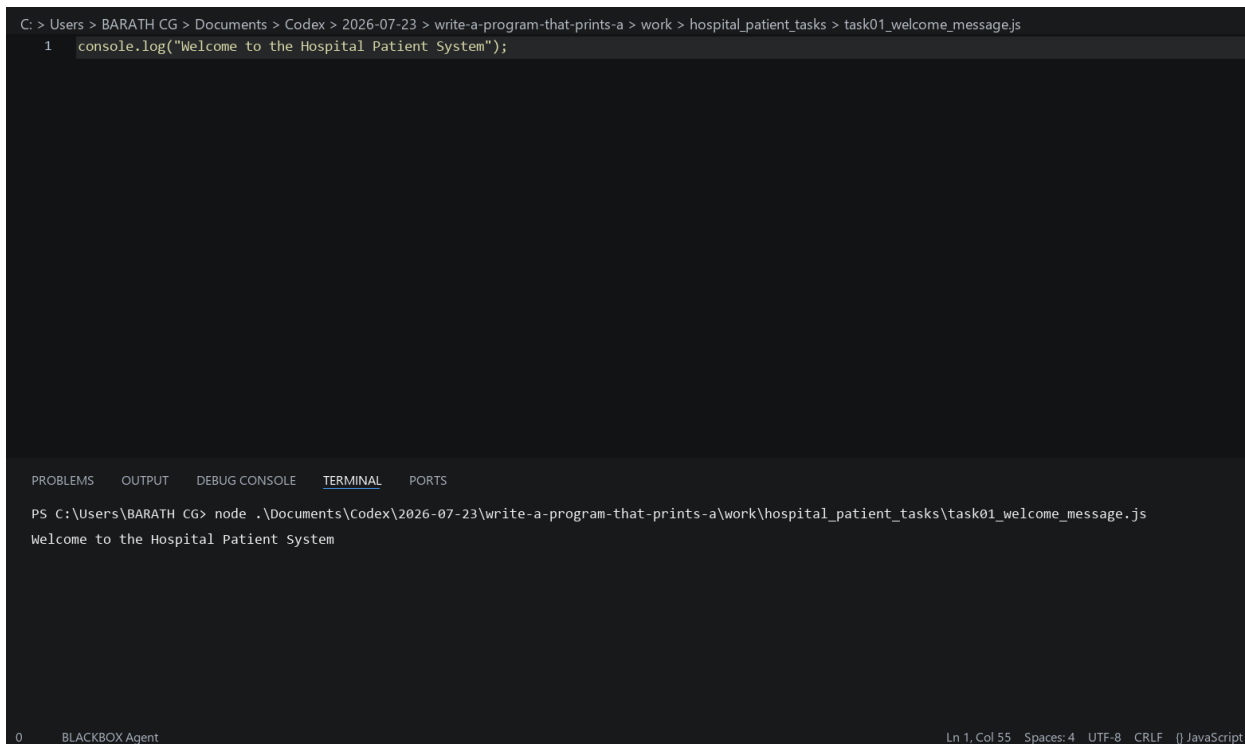
Node & Express JS CIA-1 Practical Answers - 10 Tasks

Topic	Hospital Patient System
Tasks Covered	Task 1 to Task 10
Language Used	JavaScript with Node.js

Task 1: Welcome Message

Question: Write a program that prints a welcome message for the "Hospital Patient System" to the console.

Code and Output Screenshot



```
C: > Users > BARATH CG > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > task01_welcome_message.js
1 console.log("Welcome to the Hospital Patient System");
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\BARATH CG> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task01_welcome_message.js
Welcome to the Hospital Patient System
```

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Output

Welcome to the Hospital Patient System

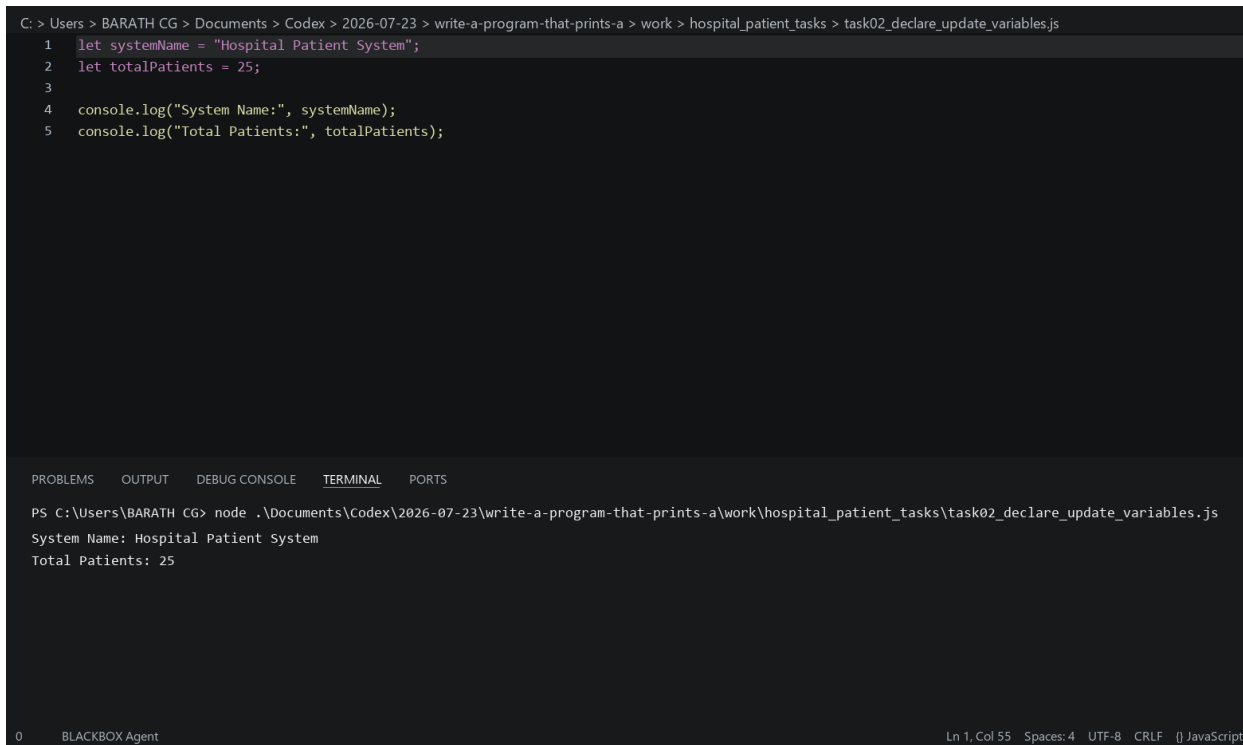
Summary: Used console.log() to display a welcome message for the Hospital Patient System.

Used: console.log()

Task 2: Declare and Update Variables

Question: Declare variables to store the system name and the total number of patients, then print both.

Code and Output Screenshot



```
C: > Users > BARATH CG > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > task02_declare_update_variables.js
1  let systemName = "Hospital Patient System";
2  let totalPatients = 25;
3
4  console.log("System Name:", systemName);
5  console.log("Total Patients:", totalPatients);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\BARATH CG> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task02_declare_update_variables.js
System Name: Hospital Patient System
Total Patients: 25
```

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Output

System Name: Hospital Patient System
Total Patients: 25

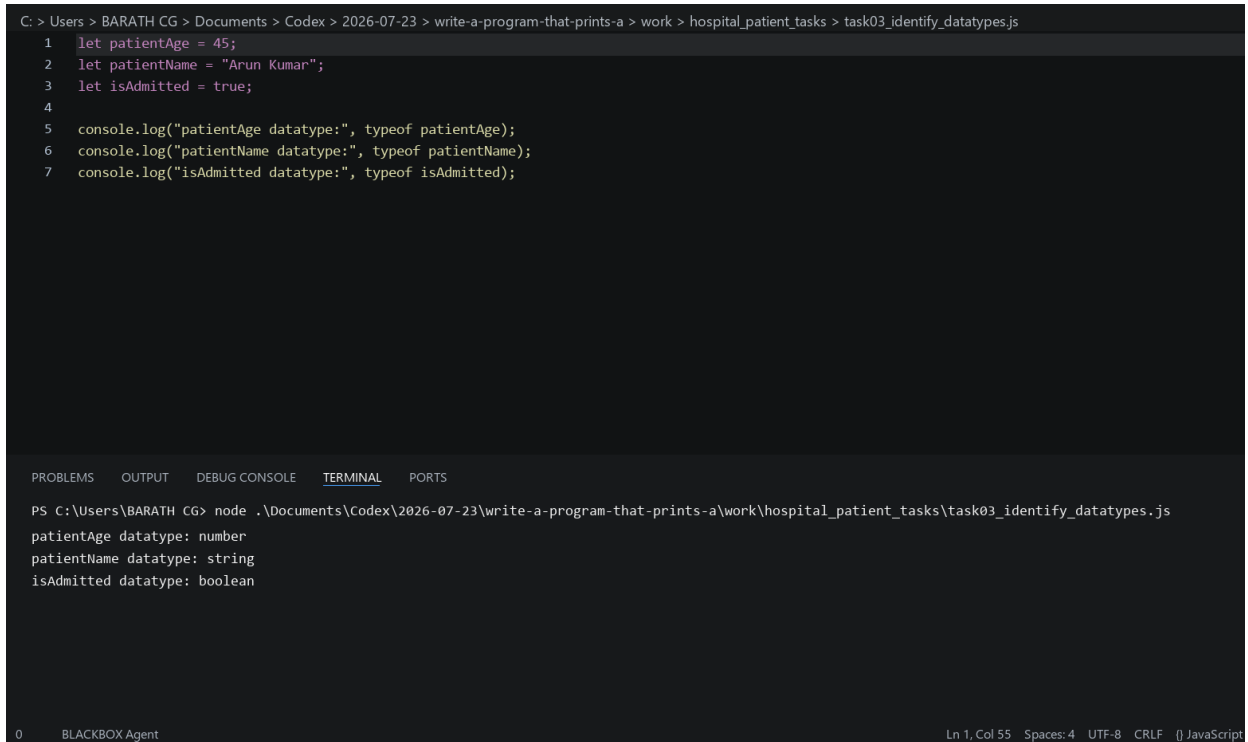
Summary: Declared a string variable for the system name and a number variable for the total patients, then printed both values.

Used: let variables and console.log()

Task 3: Identify Datatypes

Question: Declare variables of type Number, String, and Boolean related to a patient, and print their datatypes using typeof.

Code and Output Screenshot



```
C: > Users > BARATH CG > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > task03_identify_datatypes.js
1  let patientAge = 45;
2  let patientName = "Arun Kumar";
3  let isAdmitted = true;
4
5  console.log("patientAge datatype:", typeof patientAge);
6  console.log("patientName datatype:", typeof patientName);
7  console.log("isAdmitted datatype:", typeof isAdmitted);

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\BARATH CG> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task03_identify_datatypes.js
patientAge datatype: number
patientName datatype: string
isAdmitted datatype: boolean

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```

Output

```
patientAge datatype: number
patientName datatype: string
isAdmitted datatype: boolean
```

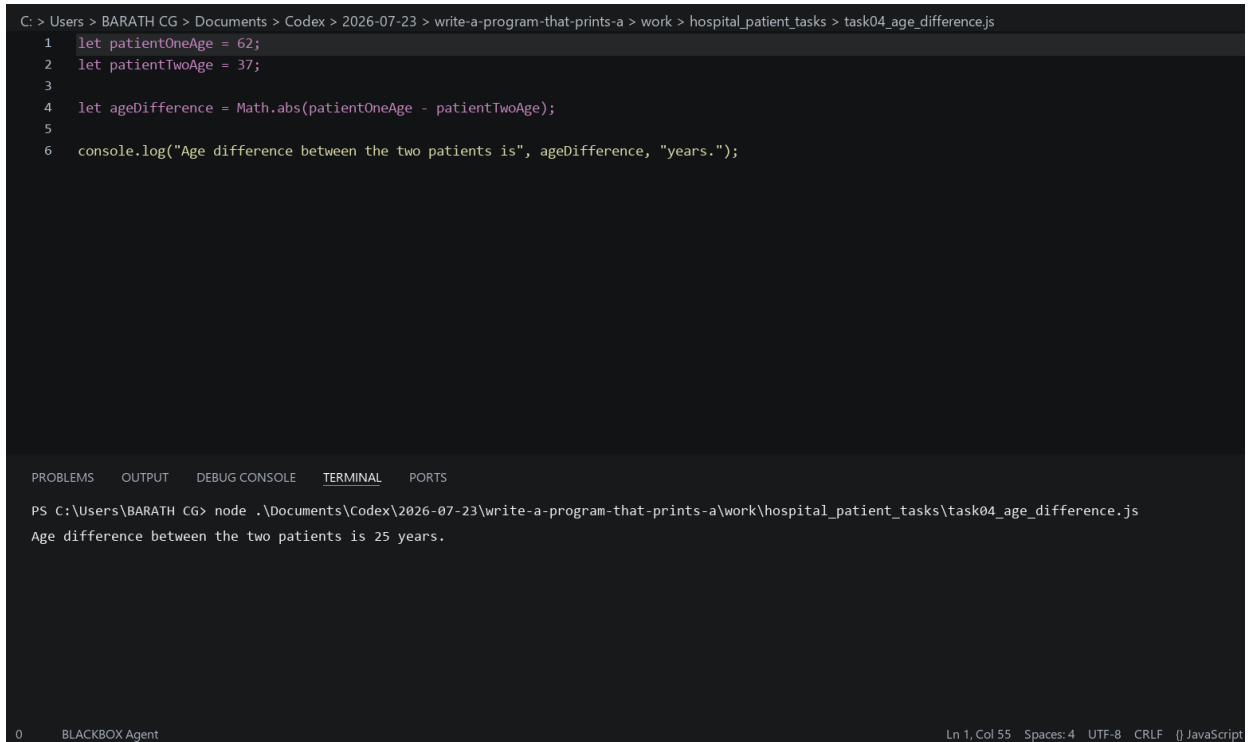
Summary: Created patient-related number, string, and boolean variables and used typeof to identify their datatypes.

Used: typeof operator

Task 4: Perform a Calculation

Question: Write a program to calculate the age difference between two patients using arithmetic operators, and print the result.

Code and Output Screenshot



```
C: > Users > BARATH CG > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > task04_age_difference.js
1  let patientOneAge = 62;
2  let patientTwoAge = 37;
3
4  let ageDifference = Math.abs(patientOneAge - patientTwoAge);
5
6  console.log("Age difference between the two patients is", ageDifference, "years.");
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\BARATH CG> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task04_age_difference.js
Age difference between the two patients is 25 years.
```

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Output

Age difference between the two patients is 25 years.

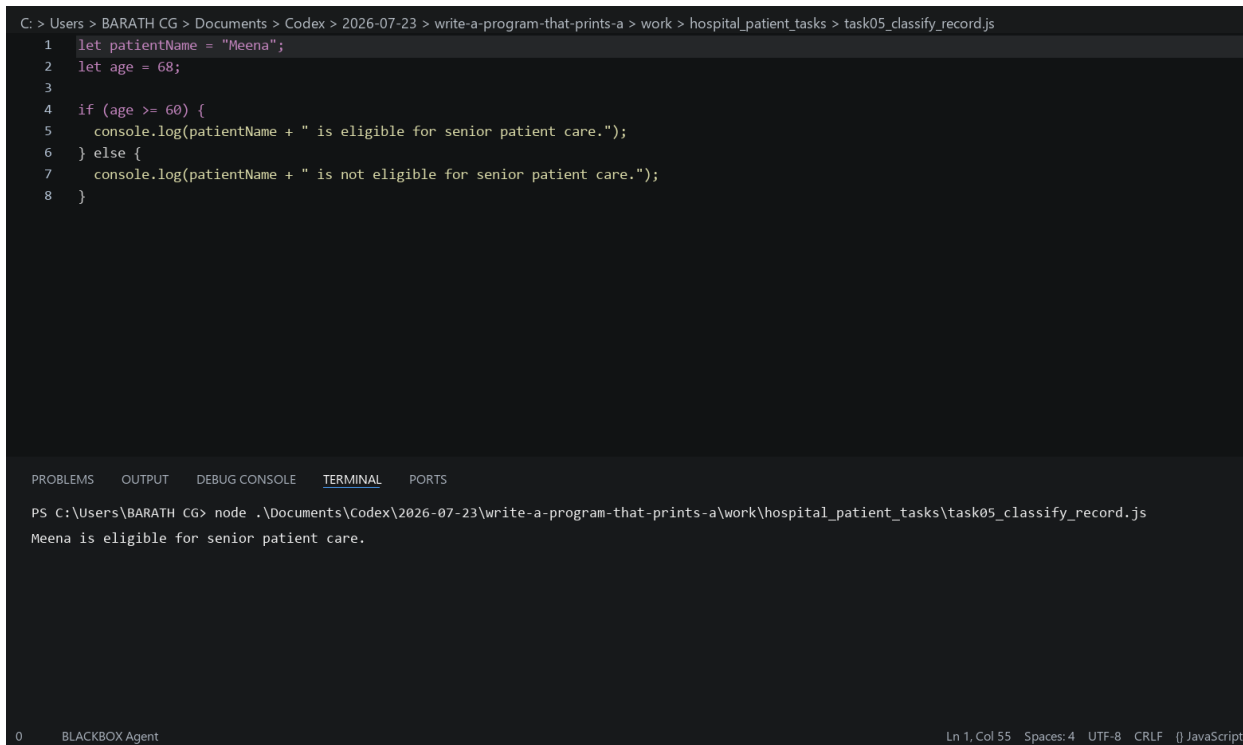
Summary: Used subtraction and Math.abs() to calculate the positive age difference between two patients.

Used: arithmetic operator and Math.abs()

Task 5: Classify a Record

Question: Write a program to check whether a given patient's age meets a specific condition using if-else.

Code and Output Screenshot



```
C: > Users > BARATH CG > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > task05_classify_record.js
1  let patientName = "Meena";
2  let age = 68;
3
4  if (age >= 60) {
5      console.log(patientName + " is eligible for senior patient care.");
6  } else {
7      console.log(patientName + " is not eligible for senior patient care.");
8  }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\BARATH CG> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task05_classify_record.js
Meena is eligible for senior patient care.
```

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Output

Meena is eligible for senior patient care.

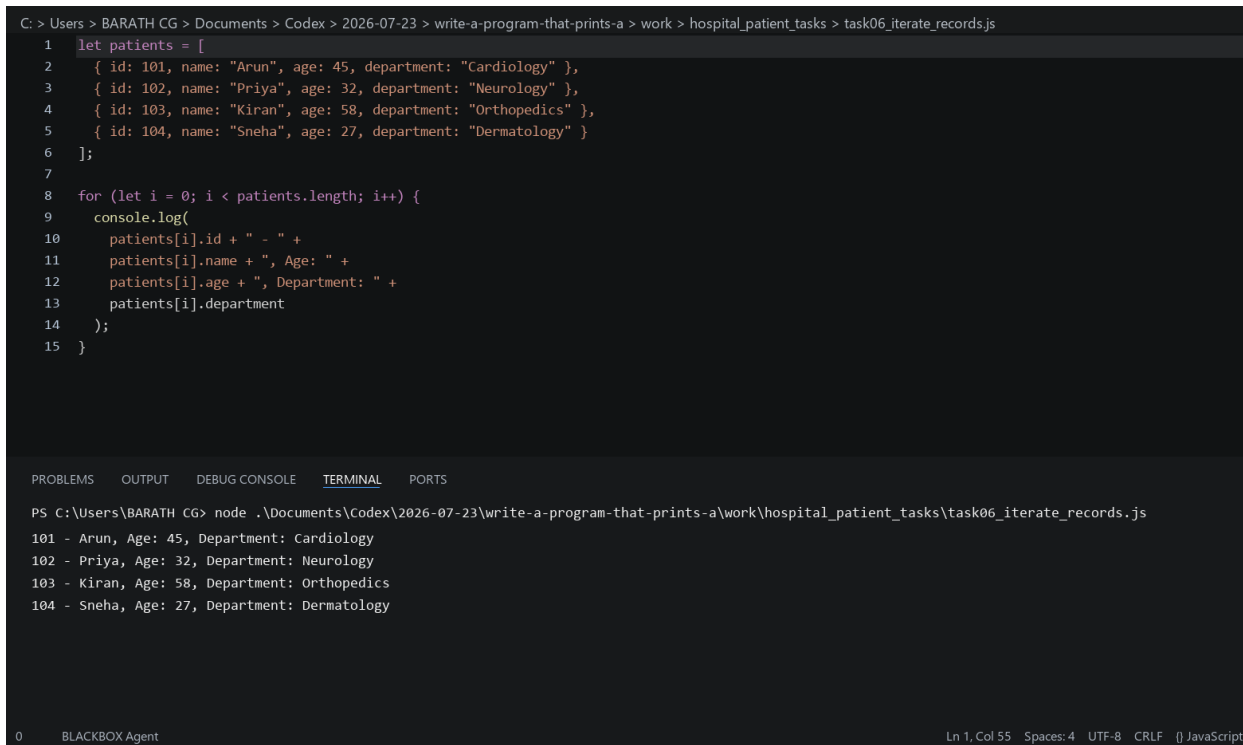
Summary: Used an if-else condition to classify whether the patient is eligible for senior care based on age.

Used: if-else condition

Task 6: Iterate Over Records

Question: Create an array of at least 4 patient objects and use a loop to print each one's details.

Code and Output Screenshot



```
C: > Users > BARATH CG > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > task06_iterate_records.js
1  let patients = [
2    { id: 101, name: "Arun", age: 45, department: "Cardiology" },
3    { id: 102, name: "Priya", age: 32, department: "Neurology" },
4    { id: 103, name: "Kiran", age: 58, department: "Orthopedics" },
5    { id: 104, name: "Sneha", age: 27, department: "Dermatology" }
6  ];
7
8  for (let i = 0; i < patients.length; i++) {
9    console.log(
10     patients[i].id + " - " +
11     patients[i].name + ", Age: " +
12     patients[i].age + ", Department: " +
13     patients[i].department
14   );
15 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\BARATH CG> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task06_iterate_records.js
101 - Arun, Age: 45, Department: Cardiology
102 - Priya, Age: 32, Department: Neurology
103 - Kiran, Age: 58, Department: Orthopedics
104 - Sneha, Age: 27, Department: Dermatology
```

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Output

```
101 - Arun, Age: 45, Department: Cardiology
102 - Priya, Age: 32, Department: Neurology
103 - Kiran, Age: 58, Department: Orthopedics
104 - Sneha, Age: 27, Department: Dermatology
```

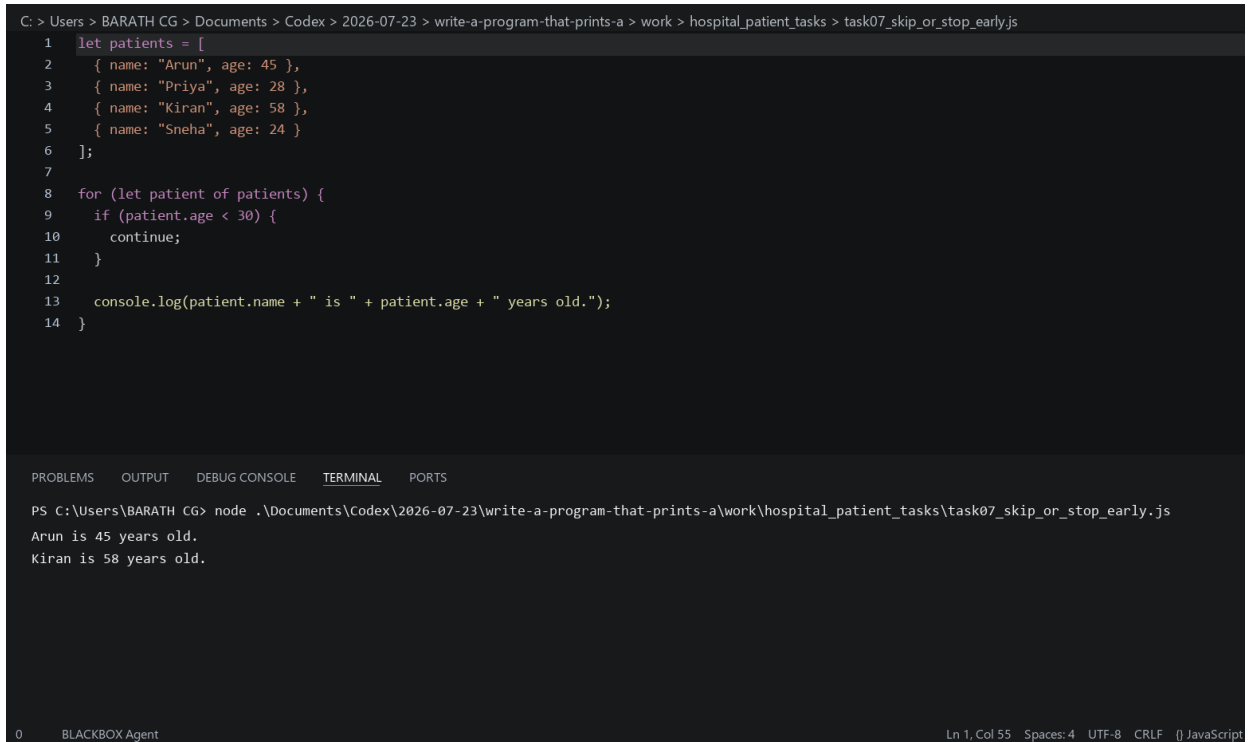
Summary: Created an array of four patient objects and used a for loop to print each patient's details.

Used: array of objects and for loop

Task 7: Skip or Stop Early

Question: Using the array of patients, use break or continue while looping to skip or stop at a specific condition on age.

Code and Output Screenshot



```
C: > Users > BARATH CG > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > task07_skip_or_stop_early.js
1  let patients = [
2    { name: "Arun", age: 45 },
3    { name: "Priya", age: 28 },
4    { name: "Kiran", age: 58 },
5    { name: "Sneha", age: 24 }
6  ];
7
8  for (let patient of patients) {
9    if (patient.age < 30) {
10     continue;
11    }
12
13    console.log(patient.name + " is " + patient.age + " years old.");
14  }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\BARATH CG> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task07_skip_or_stop_early.js
Arun is 45 years old.
Kiran is 58 years old.
```

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Output

Arun is 45 years old.
Kiran is 58 years old.

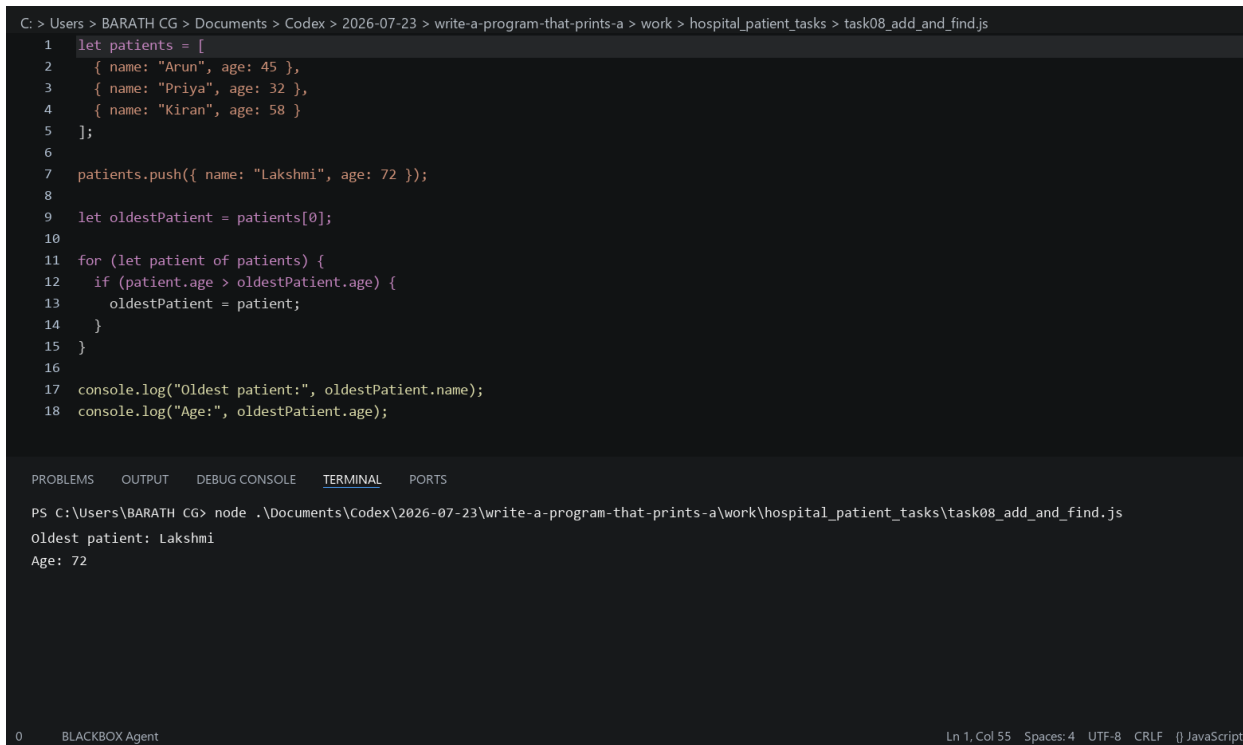
Summary: Used continue to skip patients below 30 years of age while printing the remaining records.

Used: for...of loop with continue

Task 8: Add and Find

Question: Use push() to add a new patient to the array. Write a loop to find the patient with the highest age.

Code and Output Screenshot



```
C: > Users > BARATH CG > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > task08_add_and_find.js
1  let patients = [
2    { name: "Arun", age: 45 },
3    { name: "Priya", age: 32 },
4    { name: "Kiran", age: 58 }
5  ];
6
7  patients.push({ name: "Lakshmi", age: 72 });
8
9  let oldestPatient = patients[0];
10
11 for (let patient of patients) {
12   if (patient.age > oldestPatient.age) {
13     oldestPatient = patient;
14   }
15 }
16
17 console.log("Oldest patient:", oldestPatient.name);
18 console.log("Age:", oldestPatient.age);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\BARATH CG> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task08_add_and_find.js
Oldest patient: Lakshmi
Age: 72
```

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Output

Oldest patient: Lakshmi
Age: 72

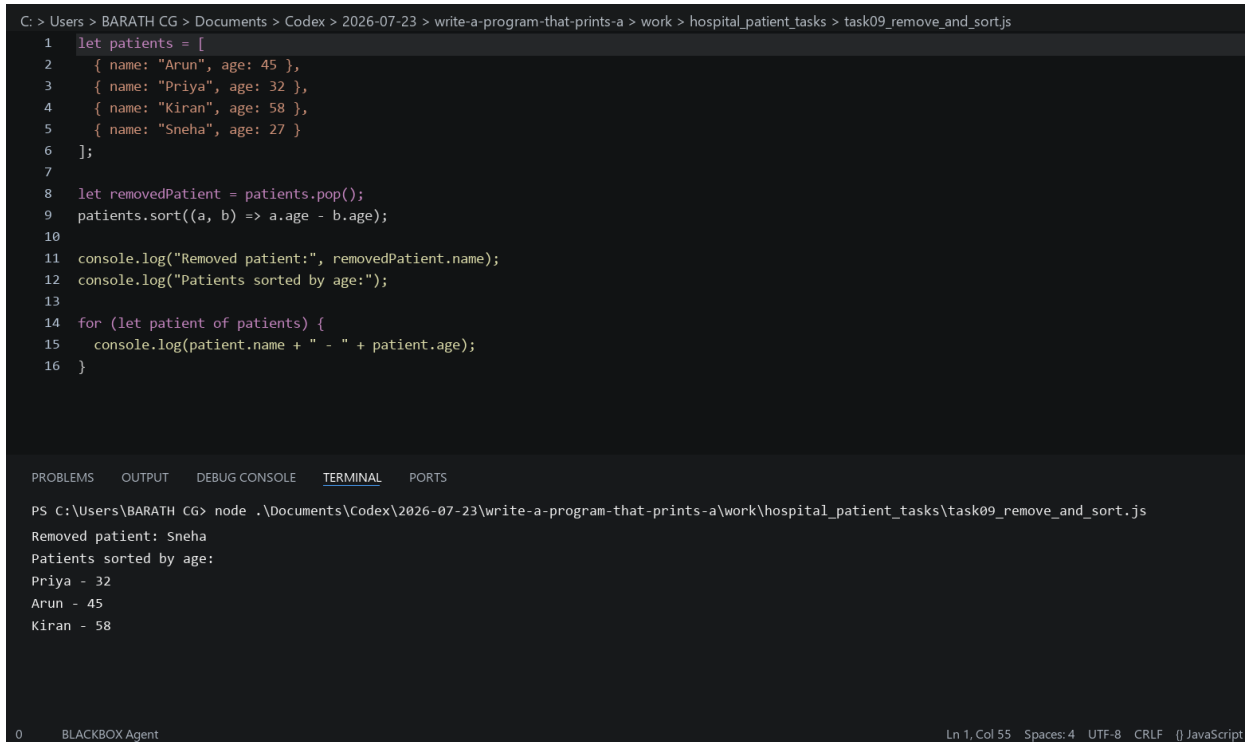
Summary: Used push() to add a new patient and a loop to find the patient with the highest age.

Used: push() method and comparison loop

Task 9: Remove and Sort

Question: Use `pop()` or `shift()` to remove an entry from the array. Use the array's `sort()` method to sort patients by age.

Code and Output Screenshot



```
C: > Users > BARATH CG > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > task09_remove_and_sort.js
1  let patients = [
2    { name: "Arun", age: 45 },
3    { name: "Priya", age: 32 },
4    { name: "Kiran", age: 58 },
5    { name: "Sneha", age: 27 }
6  ];
7
8  let removedPatient = patients.pop();
9  patients.sort((a, b) => a.age - b.age);
10
11 console.log("Removed patient:", removedPatient.name);
12 console.log("Patients sorted by age:");
13
14 for (let patient of patients) {
15   console.log(patient.name + " - " + patient.age);
16 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\BARATH CG> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task09_remove_and_sort.js
Removed patient: Sneha
Patients sorted by age:
Priya - 32
Arun - 45
Kiran - 58
```

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Output

```
Removed patient: Sneha
Patients sorted by age:
Priya - 32
Arun - 45
Kiran - 58
```

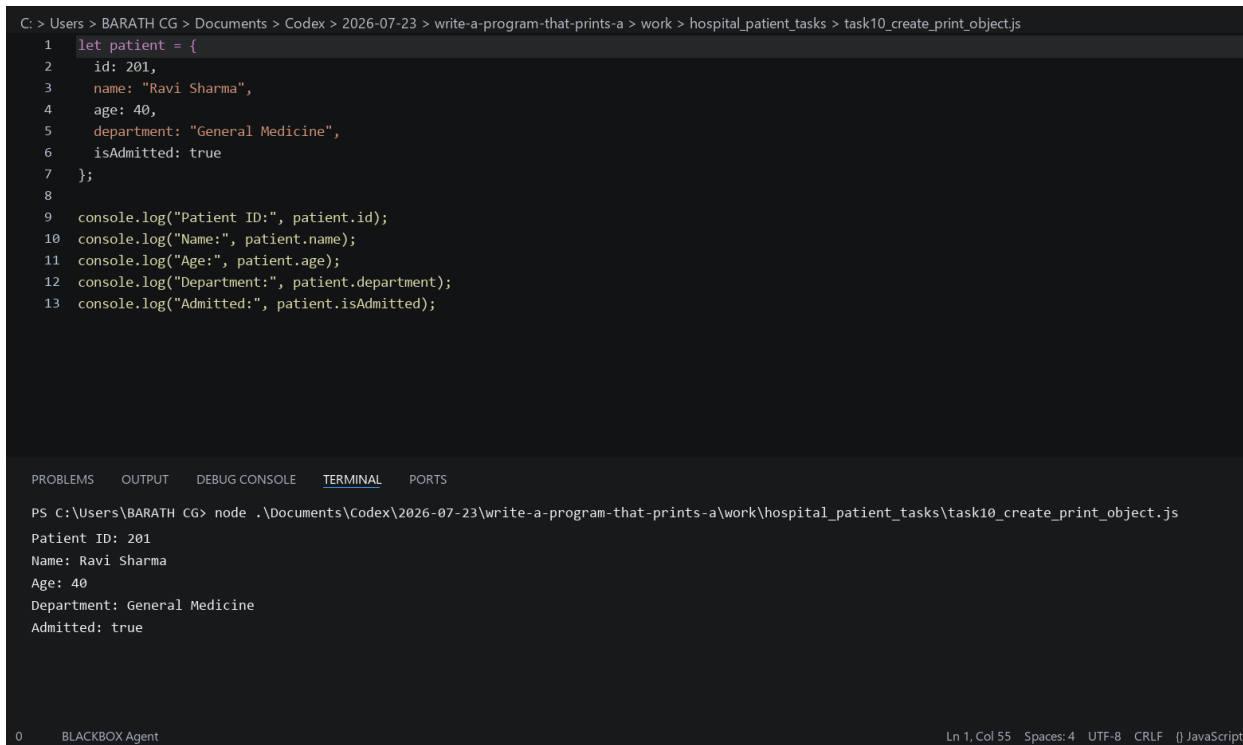
Summary: Used `pop()` to remove the last patient and `sort()` to arrange the remaining patients by age.

Used: `pop()` method and `sort()` method

Task 10: Create and Print an Object

Question: Create a patient object with at least 5 properties and print each property individually.

Code and Output Screenshot



```
C: > Users > BARATH CG > Documents > Codex > 2026-07-23 > write-a-program-that-prints-a > work > hospital_patient_tasks > task10_create_print_object.js
1  let patient = {
2    id: 201,
3    name: "Ravi Sharma",
4    age: 40,
5    department: "General Medicine",
6    isAdmitted: true
7  };
8
9  console.log("Patient ID:", patient.id);
10 console.log("Name:", patient.name);
11 console.log("Age:", patient.age);
12 console.log("Department:", patient.department);
13 console.log("Admitted:", patient.isAdmitted);

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\BARATH CG> node .\Documents\Codex\2026-07-23\write-a-program-that-prints-a\work\hospital_patient_tasks\task10_create_print_object.js
Patient ID: 201
Name: Ravi Sharma
Age: 40
Department: General Medicine
Admitted: true

0 BLACKBOX Agent Ln 1, Col 55 Spaces: 4 UTF-8 CRLF {} JavaScript
```

Output

Patient ID: 201
Name: Ravi Sharma
Age: 40
Department: General Medicine
Admitted: true

Summary: Created a patient object with five properties and printed each property separately using dot notation.

Used: object properties and dot notation